



Realistic Agent Behaviour based on Concurrent Process Architectures – Modelling Human Navigation

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Why simulating navigating people is harder than it looks

Motivation

- Navigation is the result of complex, continuous, and situated decision-making
- Urban planning, Transportation, and Cognitive Sciences demand more than crowd dynamics
- Simple models capture movement, but not the way people perceive, decide, and adapt while navigating
- Growing need for realistic human navigation models
- Challenge of modelling the concurrent and interwoven processes of human navigation



Navigation is never just one thing

Human Navigation as Concurrent Processes

Routing

- Empirical evidence: not a single behaviour, but multiple concurrent processes
 - e.g., perceiving, route planning, orienting, path following, monitoring progress

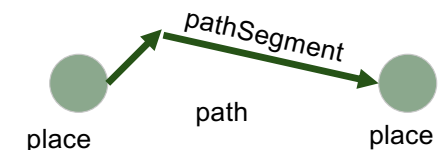
Follow path

- These processes
 - influence each other dynamically
 - happen concurrently
 - are intertwined, not independent

Orienting

- Each process has its own ontology and knowledge structure
 - Routing: topological graphs, planning the overall route
 - Path Following: places and segments, maintaining course
 - Orienting: self-localization
 - Walking: movement across surfaces

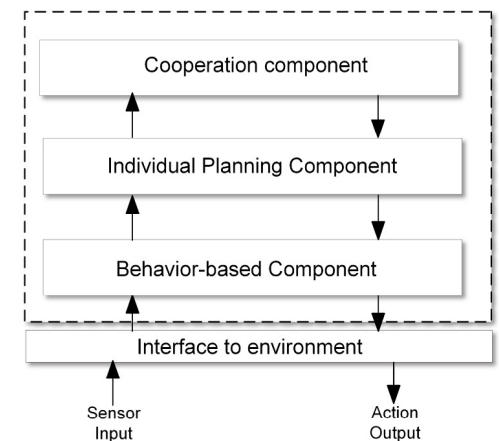
Walking



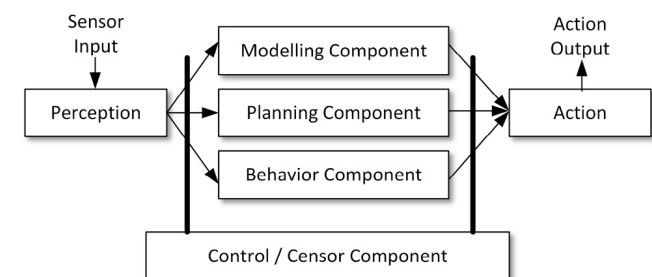
What traditional agent models get right—and wrong

Classical and generic Architectures

- An architecture provides a framework for formulating system behaviour, consisting of specific components that are responsible for specific functionalities
- Sense–Reason–Act cycle as foundation, often simplified to sequential “steps”
- Layered architectures are particularly suitable for dynamic environments and for complex tasks
 - vertical: upper layers are activated from the lower ones
 - horizontal: independent processes without interactions, additional control module needed
- Cognitive architectures (e.g., SOAR, ACT-R) cannot model locomotion
- Existing architectures explain some aspects of decision-making, but they assume sequential control where humans act concurrently



(a) Vertically Layered



(b) Horizontally Layered Architecture

If agents are to be realistic, they must multitask

Limitations of Existing Architectures

- Centralized or hierarchical control
- Weak support for true concurrency
- Over-simplified sequencing of perception, reasoning, and action
- Poor fit for intertwined navigation processes

Requirements for a New Architecture

- Concurrent reasoning processes within one agent
- Direct sensor access for each process
- Interaction via shared agent state (not strict control layers)
- Avoid imposed dependencies or suppression links
- Allow for multiple agents

Agents should think and do several things at the same time

Proposed Architecture – concurrent processes

- Multiple processes running in parallel, loosely coupled
 - Each process: specific function (walking, orienting, etc.)
 - Independent sensor access
 - Shared internal state for communication
 - Only one process issues actions (movement)
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- We introduce a concurrent process architecture where independent processes share state but act without hierarchy

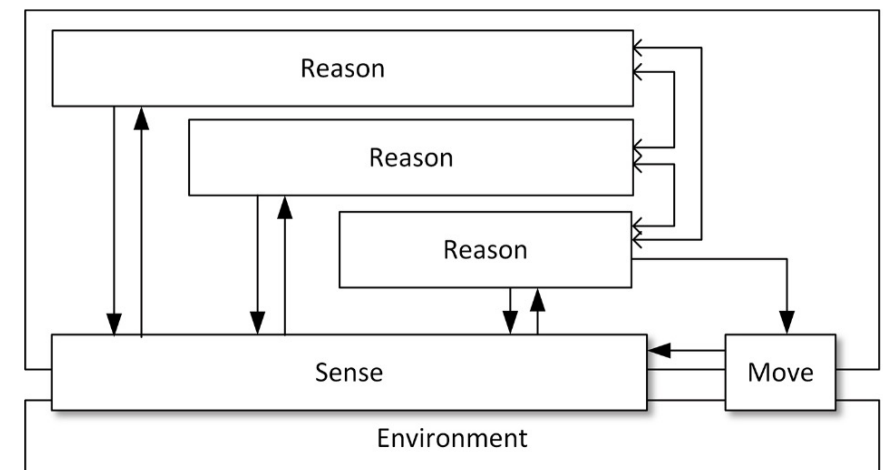
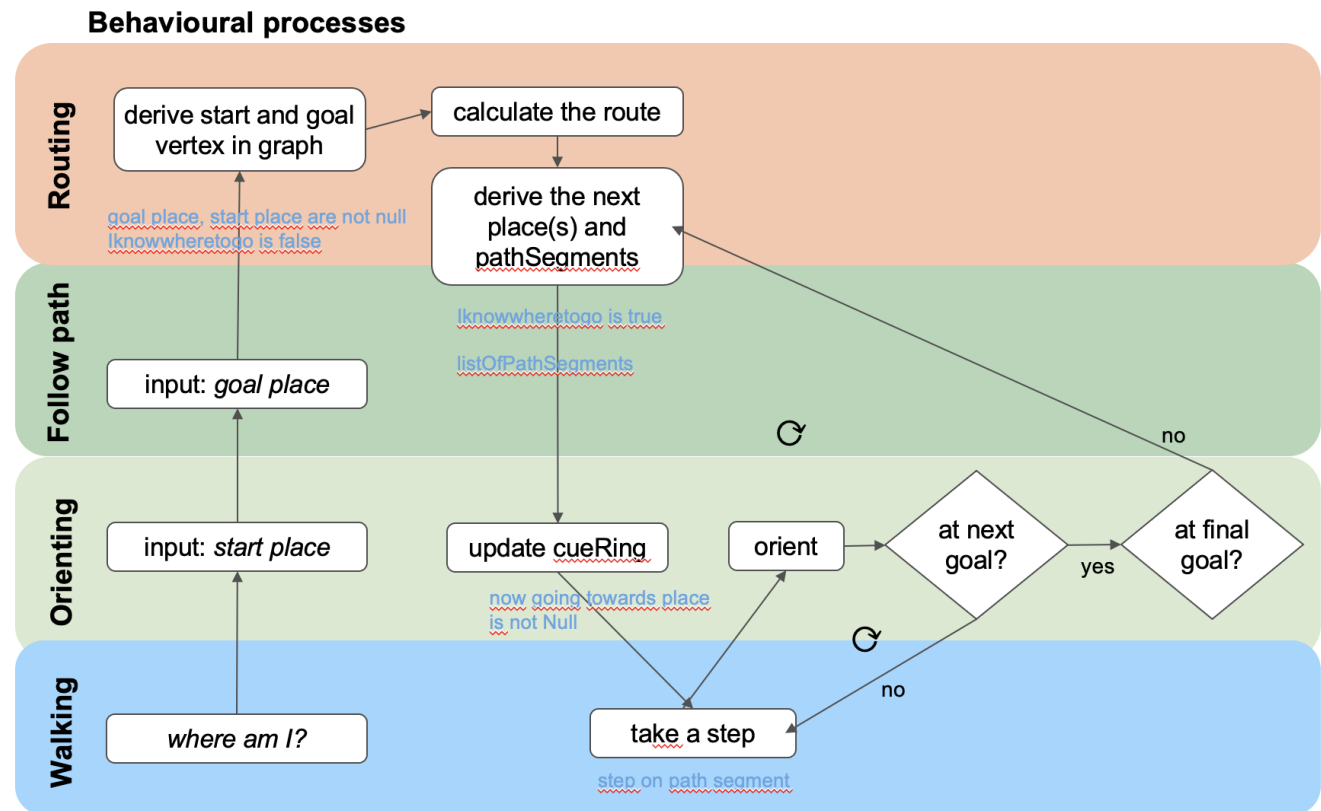


Fig. 2: Concurrent processes architecture

Four “reasoners” inside every pedestrian agent

Core Navigation Processes

- Processes stratified but concurrent
- Interaction through agent states (not hierarchy)
- Each process has its own spatial representation
- Walking, orienting, path following, and routing each run as separate processes, together producing realistic navigation



conceptual model

Putting the idea to the test on a university campus

Implementation & Example

- Four processes implemented separately: interact via shared state variables (e.g., path known, goal detected)
- Easier to separate conceptual and implementation concerns

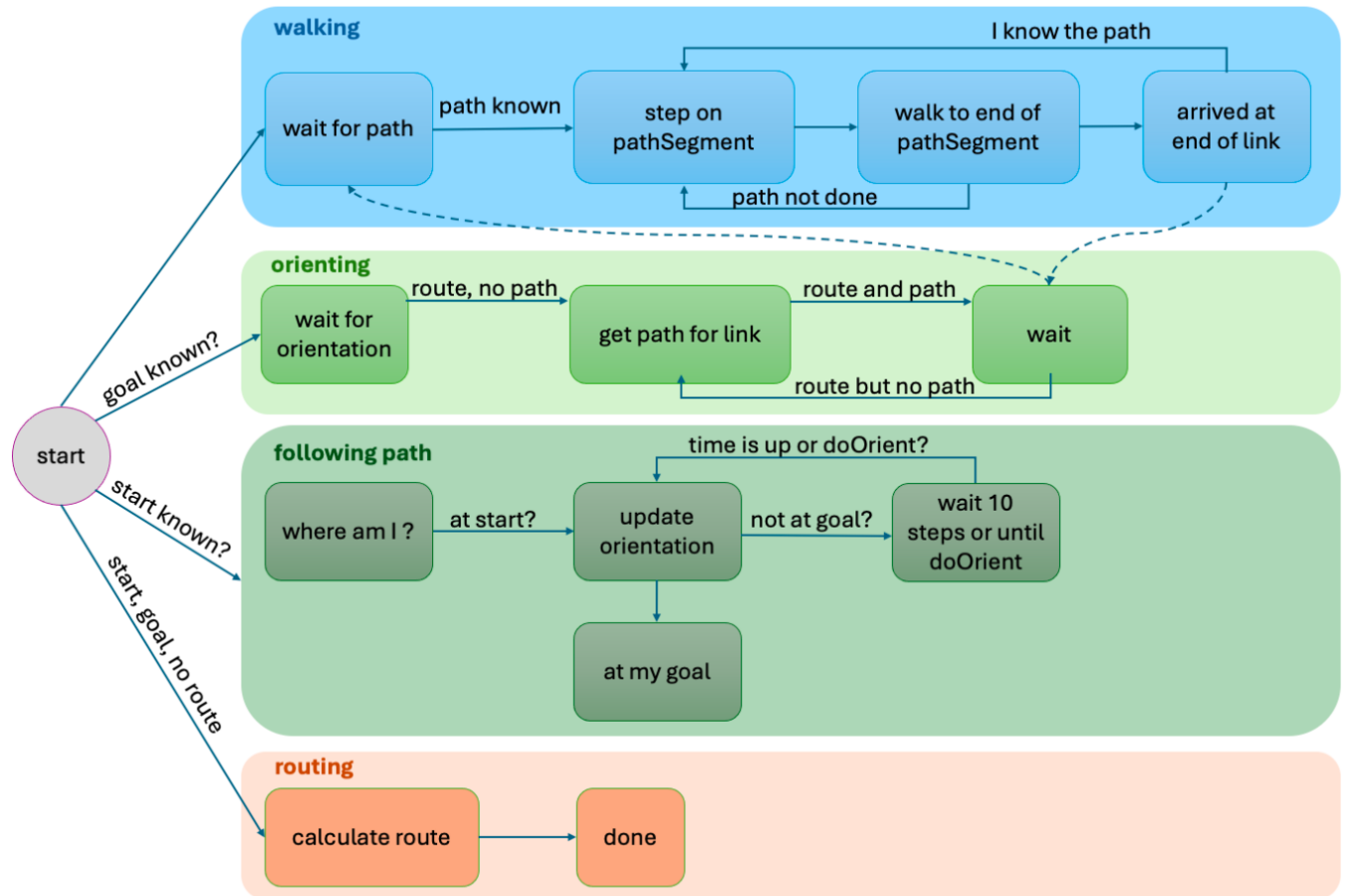


Fig. 4: Four reasoners and their interaction

Putting the idea to the test on a university campus

Implementation & Example

- Platform: SeSAM with concurrent “reasoners”
- Visual programming fits conceptual design
- Behaviour emerges from indirect interactions without central control

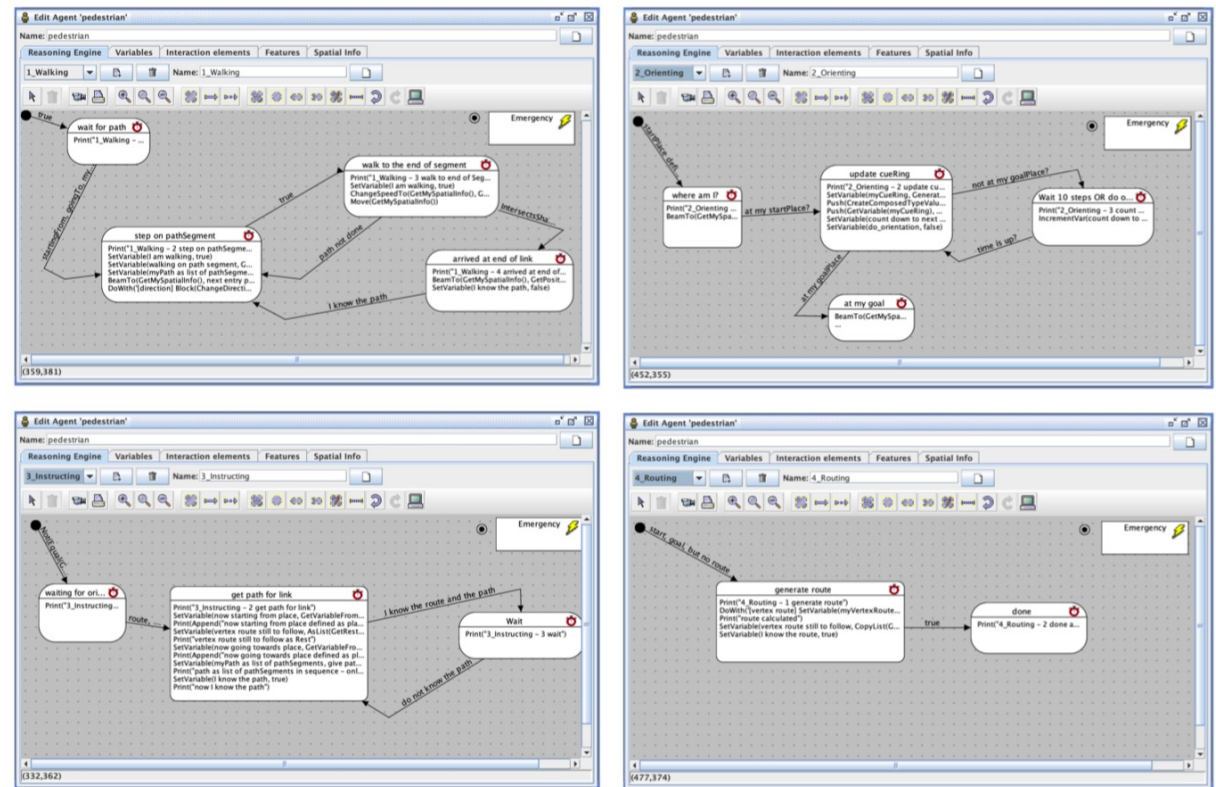


Fig. 5: Four reasoners as implemented in SeSAM

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Implementation & Example

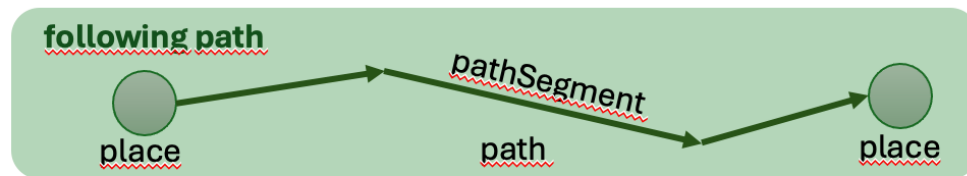
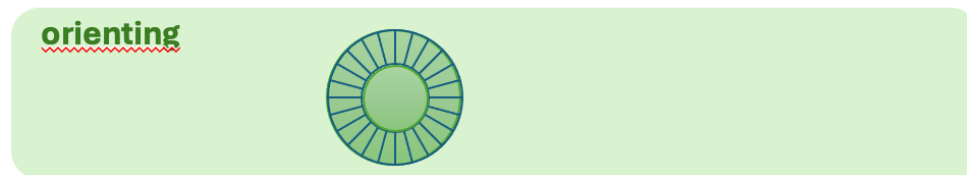
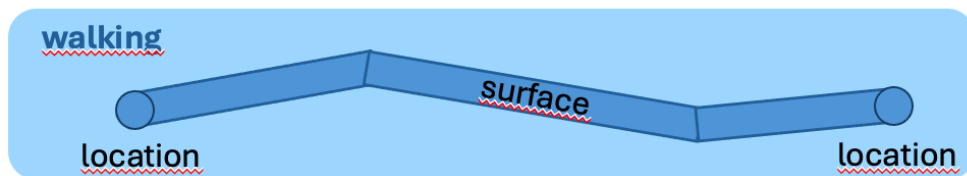
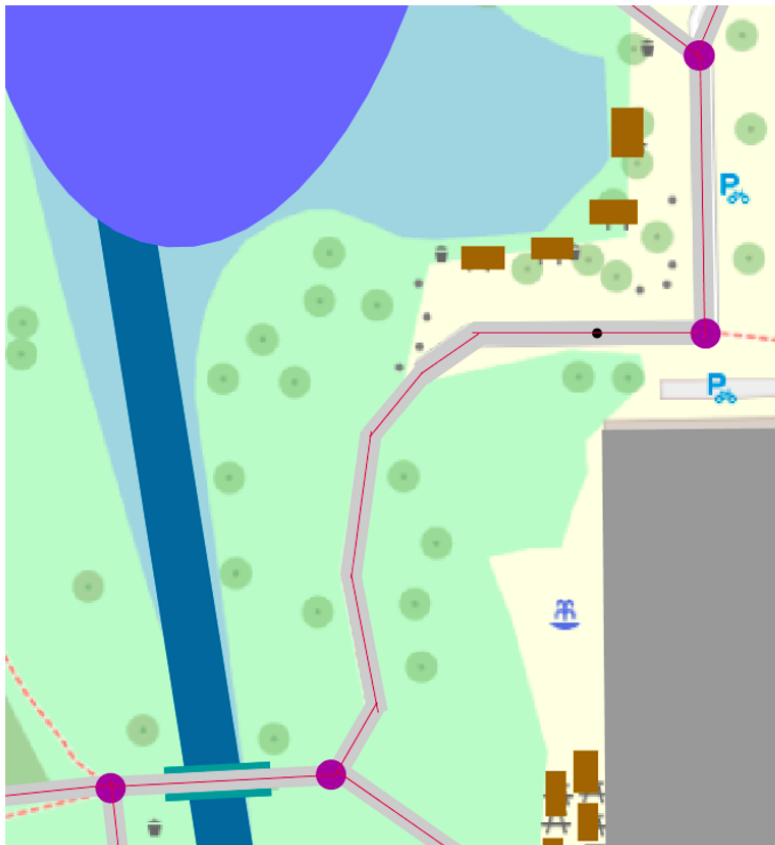
- Realistic 2D environment: map from openstreetmap.org
- Simplified geometric objects in SeSAM (on purpose)
- 50 agents crossing this part of the campus
- see also github.com/GeosavvyAgents/concurrentNavigationProcesses



CC-BY: openstreetmap contributors

Putting the idea to the test on a university campus

Implementation & Example



What worked—and what will keep us busy

Strengths and Challenges



- Stratified behavioural processes with individual spatial representations
- Reflects empirical findings in neuroscience and spatial cognition
- Modular and interpretable agents
- Avoids pitfalls of layered control (complex suppression, synchronization issues)
- Easy to extend (attention, communication, affect, error detection)
- Synchronizing the timing of concurrent processes
- Debugging and validation of the model will be more complex
- Computational overhead of multiple processes
- Still manageable with efficient scheduling

The road ahead for realistic navigation agents

Take-aways

- Concurrent processes enable realistic navigation modelling
 - Implementing each process separately makes modelling easier
 - Simple, modular, and cognitively plausible architecture
 - Framework bridges cognitive realism and computational tractability
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- Future work
 - empirical validation – compare with observed behaviour
 - richer processes – visual perception, landmark recognition, error detection, social interaction
 - large-scale integration – testing in more complex or larger environments
 - tooling support – develop tools to reduce overhead for modeller



Thank you for your attention!

Questions and comments appreciated.

